

Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

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Abstract. Courts now sanction lawyers for filing citations a language model invented, and most research responds by detecting such errors in finished briefs. This paper measures how they are produced, and what a checker does in the loop instead. Seven models from seven vendors, open-weight 30B to frontier, drafted argument sections for 48 U.S. Supreme Court matters under five conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was adjudicated deterministically, with no language-model judge: existence against an 18-million-row reporter index, quotations verbatim against opinion text with a checker validated on seeded faults and audited by hand, pinpoint pages against reporter pagination. The failure mode moves with capability. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, quotation accuracy fell for three of the seven, two of them under both prompt templates, and accurate quotations per draft fell for five on the single template that outcome was run on, two at the top of the range; the strongest model’s rate did not fall, most plausibly because it quoted less. Adjusting for the age of the cited opinion, which the date clause itself sets, removes about 45 percent of that clause’s effect on the log-odds scale and leaves the rest, so it damages integrity partly by degrading quoting and partly by moving the model onto unfamiliar authority. Fed the checker’s findings, models satisfy it by deletion: of Sonnet 5’s 35 flagged quotations 3 were corrected and 30 deleted, and accurate quotations per draft rose for one model of seven. A flag carrying the cited passage raises repair, which then overtakes deletion in three of seven models. Holding revision rounds fixed, revision with no feedback interpolates to a nominal precision between 0.56 and 0.66, or 0.44 to 0.52 realised, the point below which a verifier destroys more correct work than leaving the model to revise alone. Both experiments make one point: a rate computed over what a model chooses to expose to a check can rise while the correct output behind it does not, so an evaluation should report the count beside the rate.

Keywords. legal citation, hallucination, large language models, verification, evaluation

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1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs with citations a language model invented, from *Mata v. Avianca* [1] to a docket a public tracker now counts in the thousands [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model find [3,4]? That is the right measure for reviewing a filed brief, not for asking which prompting conditions produce the errors, this paper’s question.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and for which models? Deployed prompts carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a publication-year window lowered citation existence even in capable models; we add a sanctions warning and know of no comparable study for law. Second, when a deterministic checker feeds its findings back into revision, does the model correct them, leave them, or change the draft so that the tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the highest-recall published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), which its authors attribute to page-level information living behind commercial databases [3]. We built it from public sources: existence against a local index from CourtListener bulk data, verbatim quotation checking validated on seeded faults and audited on real drafts, and page-level pinpoint verification for quotations on U.S. Reports authority against the official pagination in the Caselaw Access Project’s archives [6].

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings to date [2]; a grounded arm with ten retrieved authorities in the prompt measures what retrieval changes. Drafts under the four non-baseline conditions are the pressured drafts (1,344 here, three refusals). The paper contributes a constraint-sensitive evaluation, in which reliability at a neutral prompt does not predict reliability under drafting constraints; deterministic verification of existence, quotation, and pinpoint page with no language-model judge, released with the drafts; and the finding that verification is not repair, since flags fed back to the model raise measured accuracy by deletion and an imprecise verifier damages correct content.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their categories and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for a named case’s citation almost never produce it; we ask whether the cases a model chooses to cite exist. Dahl et al. [7] and Magesh et al. [8] measure hallucinated authority in chatbots and research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified against two bibliographic services, which left 36 to 61 percent of citations unresolved. We transplant their year-window and breadth regimes to law against a full-text index that leaves 0.2 percent unresolved, and add a statistical model.

Revision with execution feedback. The loop follows execution-feedback work in code generation [9], with its no-feedback control separating feedback content from extra revision; we add controls in which a quarter to all of the feedback is false.

AI and law. Blair-Stanek and Van Durme [10] found frontier models flipping the winning party across identical zero-temperature runs, which Section 8 bounds with repeated generations. Evaluation here pairs automated metrics with the checks lawyers perform [11,12]; citation integrity is one with deterministic ground truth. Garneau et al. [13] found retrieving the paragraph a proposition rests on harder than retrieving the case. Zhang et al. [14] show models miss overruled precedent, which our date clause makes more likely and our checker does not test; ConsRAG [15] constrains retrieval, and Trozze et al. [16] and Harasta et al. [17] study how readers perceive model-written documents. Evaluations of generated legal argument [18,19] do not check the authority it cites, and citation-network work [20,21,22,23] studies the network our checker queries.

3. Tasks, Conditions, and Instruments

Terms. In 347 U.S. 483, 495, the second page is the *pincite*, for a quotation the page holding the quoted language. For a draft d let A_d be its *adjudicable* citations, labelled exists or not found, and S_d its *scored* quotations, those it attributes to a case whose text the checker holds. The outcomes are

$$E_d = \frac{|\{c \in A_d : \text{exists}\}|}{|A_d|}, \quad R_d = \frac{|\{q \in S_d : \text{accurate}\}|}{|S_d|}, \quad K_d = |\{q \in S_d : \text{accurate}\}|,$$

citation existence, *strict quotation accuracy*, and the *count* of accurate quotations in the draft; the lenient rate R_d^ℓ counts near misses as accurate. A quotation the draft does not attribute is outside S_d and so outside R_d .

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020 drawn from the Supreme Court Database [24]; each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, the Court’s decision excluded. The model argues an assigned side in under 900 words; nothing here predicts an outcome, so the memorisation confound documented for classifying Supreme Court cases does not arise [25]. A second task uses forty-eight published F.3d decisions from the Caselaw Access Project archive (1995 to 2019, four from each of the First through Eleventh and Federal Circuits, no cross-appeals); the packet is the opinion’s statement of the case and facts, cut before the analysis and scrubbed of outcome sentences, argued for the appellant under all five conditions.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”), a clause no user writes, a checkable stand-in for the authority limits practitioners impose; unlike a jurisdiction limit, which pushes toward the circuit authority where the second task finds fabrication, it moves citations toward well-memorized opinions; stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses, so a combined effect cannot be assigned to one of them. Compliance with it is partial and uneven, 0 to 35 percent of checked citations violating it with no ordering between violation and effect, so contrasts are intention-to-treat across models

receiving different doses. Each condition has one fixed template, at temperature 0. The stakes clause names Rule 11, invoked in the sanctioned-filing cases [1], which governs district-court civil actions [26] and neither task’s forum, so it is a stakes signal only. A second template rephrasing every condition was run on all 48 matters. A *grounded* arm reran baseline and combined with the ten U.S. Reports opinions most similar to the question and facts in the prompt (TF-IDF over 25,250 cached opinions decided before argument and, for combined, before 1970, neither party named), each with citation, year, and its best-matching 250-word passage, quotation restricted to the excerpts.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens (identifiers in the repository). Tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are the top of the range. An eighth model, GLM-4.7-flash, exhausted a 30,000-token budget in reasoning on 33 of 240 drafts and is excluded.

Adjudication. Full case citations are extracted with *eyecite* [27] (statutes and rules cite no case and are outside every rate) and labelled exists, not found, or unverifiable, the last for a reporter the index does not carry, so coverage is never counted as fabrication; 0.2 percent of the 14,040 records are unverifiable. Citations resolve against an 18-million-row table; quotations are located by balanced quotation marks, which block quotations lack, though that is no route out of the scored set here (12 block-quoted lines in 1,680 drafts, none Sonnet 5’s); they are attributed, in order, to the citation whose parenthetical holds them, the case named in their own sentence, the citation following them in that sentence (not one introduced by *see* or *cf.*), the citation an *Id.* points to, or the last citation in the paragraph; one introduced by a statute, rule, the record, or a lower court, or with no citation in its paragraph, is left unpaired, the checker having nothing to test it against: 37 percent of the 10,846 quotations in the full run: 24 percent introduced by non-case material, 10 percent in a paragraph that cites nothing, 3 percent too far from any citation. *Exists* means the volume, reporter, and page resolve to an indexed opinion (13,241 of the 14,040 records; 774 are not found); for the 11,769 whose draft names the parties the name is compared separately, and 205 (1.7 percent) carry a name the index does not match, abbreviated names included; counting every mismatch as not found keeps the 3 surviving existence contrasts, adds Mistral Small’s temporal fall (corrected $p=0.046$), and removes Sonnet 5’s uncorrected combined rise. A quotation q is split into fragments $f_1, \dots, f_n$ at ellipses and bracketed alterations (a lawful substitution is read as an omission and the words around it must match), alteration parentheticals dropped, and compared with the normalized text T of the case it is attributed to: *accurate* when every f_j is a substring of T , *near miss* when $\min_j \text{cov}(f_j, T) \geq 0.85$, for cov the share of a fragment’s words of five or more letters occurring in T , and *inaccurate* otherwise. Matching is case and punctuation insensitive, and a quotation that drops an internal citation without leaving an ellipsis, which the citation rules permit under a parenthetical, passes when the opinion holds the fragment’s halves in order across a citation-shaped gap; the lenient rate is a diagnostic, since 0.85 coverage admits one wrong word in seven. The quotation checker scores 280 of 280 on twenty synthetic drafts planting fourteen item kinds, six attribution traps and two omissions without an ellipsis among them, and the existence and pinpoint checks 100 of 100. Pinpoint pages are verified against the official pagination in the Caselaw Access Project’s HTML volumes, which mark every page break; a *pincite* naming a range or list

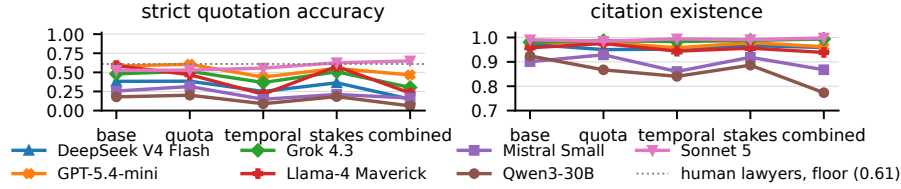


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model; the dotted line is the human reference of Section 3, a severity floor on OCR-converted briefs, not a comparison; the right axis starts at 0.7.

is credited on any page it names, a quotation is located by its longest fragment, and only those whose fragment is somewhere in the cited case enter the page-level rate.

Human reference. The same instrument scored 482 pre-ChatGPT appellate brief excerpts from Liu et al. [3] (every brief-derived excerpt of theirs carrying no injected error, their model-written subset excluded; OCR-converted, so a floor); human lawyers reach 0.610 strict and 0.788 lenient quotation accuracy (410 scored) and 0.957 existence; crediting the 57 failures that are near misses without any alteration, the signature of a corrupted character, lifts the floor to 0.749. It is a severity reference only, since human briefs differ in citation mix and the remaining failures are unaudited. The literal matcher the strict standard replaced, which read a bracketed alteration as its contents, puts the human rate at 0.507 and moves model cells by at most 9 points, the same 8 contrasts surviving.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for the existence or accuracy indicator y_{imk} of item i in matter m under condition k , exchangeable within matters [28], so the 48 matters are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model, the family the paper draws conclusions about, and call an effect significant only when it survives; counts of models are descriptive, and the effective sample is 48 matters. Every other fit is exploratory and follows the same model and the same rule, Holm within model over that fit’s own tests: the lenient outcome, distinct authorities, the second template, and pooled templates, tasks, and generations (Section 8), each pooling with its fixed effect; the released matrix gives every contrast under every fit. For the revision experiment, pre-registered descriptively, we add matter-paired comparisons between arms (Wilcoxon, bootstrap interval). Hypotheses and decision rules were frozen before the full run; every later decision, including two withdrawn findings, is in a dated ledger.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. Only the 30B model’s existence moves significantly, 0.924 to 0.773 under the combined condition, with the quota, temporal, and combined contrasts all surviving correction (combined OR 0.30, CI 0.18 to 0.51, corrected $p < 0.001$). Mistral Small falls without reaching significance and the other five stay between 0.93 and 1.00 everywhere, on cells as small as 3 against 1 not found, so those nulls are failures

Table 1. Baseline versus the combined condition; quotation columns give the strict rate over attributed quotations (lenient) [scored], existence columns the rate [not found / adjudicable], the last the share of checked citations decided after 1969 under the temporal condition; an asterisk marks a contrast surviving Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Viol., temporal
Qwen3-30B	0.179 (0.607) [229]	0.061 (0.322) [326]*	0.924 [28/370]	0.773 [90/397]*	20%
Mistral	0.255 (0.620) [184]	0.164 (0.455) [244]	0.901 [37/373]	0.868 [69/521]	35%
DeepSeek	0.383 (0.733) [251]	0.149 (0.491) [281]*	0.972 [10/354]	0.959 [20/488]	11%
Grok 4.3	0.480 (0.757) [152]	0.302 (0.647) [119]	0.981 [5/257]	0.994 [3/479]	0%
Sonnet 5	0.527 (0.802) [258]	0.649 (0.820) [194]	0.991 [3/318]	0.998 [1/450]	0%
GPT-5.4-mini	0.583 (0.819) [127]	0.466 (0.693) [88]	0.981 [8/417]	0.962 [27/713]	13%
Llama-4	0.588 (0.784) [51]	0.230 (0.557) [61]*	0.958 [9/213]	0.939 [31/506]	10%

to reject, not evidence of no effect. Each rests on at least 28 citations not found per arm, a record being one occurrence of a full citation; refitting on distinct authorities per draft leaves the same 8 contrasts.

Quotation accuracy. The temporal restriction damages quoting most; after correction it falls under temporal and combined pressure for Llama-4 (0.588 to 0.230) and Qwen3-30B (0.179 to 0.061) and under the combined condition for DeepSeek; the other three fall in the same direction without surviving correction, and Sonnet 5 does not fall (below). Llama-4 shows the largest drop in this fit, 0.588 to 0.214 under the temporal restriction (OR 0.18, CI 0.08 to 0.43, corrected $p < 0.001$) on cells of 51 and 42, and it does not replicate under the second template; on the lenient outcome 7 contrasts survive. A fall under the date clause could be damage to quoting or a change in what is quoted, since it moves the median year of Sonnet 5’s cited cases from 1986 to 1943. The clause sets that year, so it is a mediator, not a confounder: adjusting for it decomposes the effect rather than removing a bias, and odds ratios are not collapsible, so part of the movement is mechanical. Pooled over models with a model effect on the 3,644 dated quotations, the temporal clause’s odds ratio moves from 0.62 to 0.77 ($p=0.035$) once decision year enters the fit and the combined condition’s from 0.56 to 0.68 (CI 0.53 to 0.88, $p=0.003$), removing 45 and 35 percent of the log-odds effect respectively. The assumption-light alternative disagrees: inside a pre-1970 bucket, which still leaves the arms twelve years apart in median, neither pooled contrast survives correction (combined Holm 0.099). Descriptively, 24 percent of the temporal arm’s quotations cite authority the model also cites at baseline and score 0.409 against the baseline’s 0.397, the rest 0.252; that split also conditions on a choice the treatment causes. With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.21 to 0.53), Sonnet 5 excepted, and Mistral Small’s existence fall also survives (OR 0.62, corrected $p=0.004$).

Accurate quotations per draft. The rate R_d is conditional on quoting: scored quotations per cell run from 42 to 326 and the unpaired share from 15 to 50 percent, moving with the treatment. We therefore also test, post hoc and on the first template only, the count K_d (Wilcoxon paired by matter, $n=48$, Holm within model). The count falls from baseline to combined for all seven models and survives correction for 8 contrasts in five models, all under the temporal or combined clause: Qwen3-30B, DeepSeek (2.00 to 0.88 per draft), Llama-4 (0.62 to 0.29), and two whose rate contrasts did not survive, GPT-5.4-mini (corrected $p=0.033$) and Grok 4.3 ($p=0.011$). Sonnet 5’s count is flat (2.83 to

2.62, $p=0.579$) while its failing quotations per draft, near misses and inaccurate together, fall from 2.54 to 1.42, the evidence for selection below.

Robustness to phrasing. Under the second template 12 contrasts survive, 5 of them among the 8 above: Qwen3-30B’s four and DeepSeek’s combined quotation fall, the only effects holding under both templates, so Llama-4’s two quotation falls and Qwen3-30B’s quota existence fall are single-template results. Pooling both, all 8 survive and the combined quotation fall survives for 6 of seven models, Sonnet 5 excepted.

Second task and retrieval. Baseline existence on the 48 appellate matters runs from 0.652 for Qwen3-30B to 0.993 for Sonnet 5, lower than on the Supreme Court matters for 6 of seven, and concentrates in the federal reporters, 22 percent against 4 percent for the U.S. Reports; the index resolves all but 3 of the 759 federal-reporter citations in the deciding opinions themselves, so the shortfall is not coverage. A grounded arm with ten verified authorities in the prompt cut citations not found from 341 in 5,856 to 74 in 5,605 and raised quotation accuracy in every cell, yet no cell exceeded 0.753 with the source passages there, and 60 percent of Sonnet 5’s 63 remaining inaccurate quotations are paraphrases. One pressure contrast survives within the task, Llama-4’s existence fall under the temporal clause (OR 0.32, corrected $p=0.018$). Pooling both tasks, 7 of the 8 survivors remain, joined by 5 further falls and Sonnet 5’s rate rise under the stakes clause (OR 1.59, corrected $p=0.009$).

The strongest model does not degrade; the leading explanation is that it quotes less. Sonnet 5’s strict rate is numerically highest under the combined condition (0.527 to 0.649; uncorrected $p = 0.0251$, corrected $p = 0.201$), a rise that survives no fit. It wrote 9.6 quotations per draft at baseline and 6.1 under the combined condition, so the quotations it stopped writing were disproportionately the ones that would have failed. Selection is the leading reading; which were dropped, by length or case age, is not tested, and the temporal restriction also shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986). With three generations pooled Sonnet 5’s existence rise survives the logistic fit (corrected $p=0.006$) but rests on twelve citations not found (7 of 975 against 5 of 1,324), on which an exact test gives $p=0.38$; the quotation rise reaches corrected $p=0.367$ there. In the appellate task the stakes clause alone raises the rate (0.475 to 0.624, corrected $p=0.057$) without moving the cited cases (median year 1991 in both); the direction is consistent across tasks.

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop: each round the checker returned one line per failure, the model revised, and the loop ended on a clean draft or after three revisions (a round is a draft the checker read, so an episode reaching the cap has four). A *false feedback* control gave the same number of identical lines aimed at correct items, *mixed* controls replaced each line with a false one at random so that the feedback’s nominal precision π , the share of its lines drawn from the checker, was 0.25, 0.5, or 0.75 ($\pi = 1$ under true and $\pi = 0$ under false feedback), replacing rather than adding lines, so recall falls with precision and the arms vary both. A *no-feedback* control gave the same rounds with a generic instruction to revise. The loop used the first 24 matters; rates are means over drafts with at least one scored quotation (68 of the 168 true-arm episodes ended with none, two without passing the checker, 19 of Grok 4.3’s 24, so its final rate

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft. corr.: an accurate quotation on the same citation stands in its place; left: still flagged; removed: deleted, dequoted, or made inaccurate; added: new, accurate/flagged; strict accuracy as the draft mean, pooled in parentheses, round 0 / final; Clean: passing draft, with quotations / with none.

Model	Flagged at round 0					Accurate at round 0			Strict, r0 → final (pooled)	Clean
	n	corr.	dequoted	deleted	left	n	removed	added		
Qwen3-30B	139	0	10	13	116	12	0	0/32	0.089 → 0.092 (0.079/0.075)	0/0
Mistral	92	4	4	40	44	14	1	1/27	0.238 → 0.354 (0.132/0.202)	3/7
DeepSeek	134	2	58	71	3	31	5	1/2	0.301 → 0.958 (0.188/0.853)	11/12
Grok 4.3	37	0	23	14	0	13	3	0/0	0.234 → 1.000 (0.260/1.000)	5/19
Sonnet 5	35	3	2	30	0	69	4	0/0	0.743 → 1.000 (0.663/1.000)	23/1
GPT-5.4-mini	22	1	1	19	1	31	5	0/1	0.538 → 0.857 (0.585/0.931)	11/9
Llama-4	33	0	5	28	0	11	2	0/0	0.273 → 1.000 (0.250/1.000)	6/18

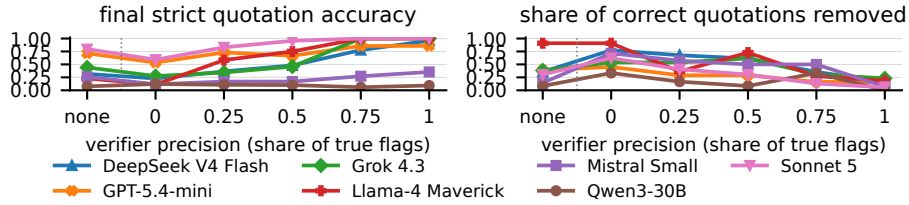


Figure 2. Nominal verifier precision π (share of true feedback lines; leftmost point, no feedback) against final strict quotation accuracy (left) and the share of accurate round-0 quotations that revision deleted, dequoted, or made inaccurate (right).

rests on 5 drafts and Llama-4’s on 6). The loop was rerun after the attribution rule was corrected; the original run, whose true-arm flags had realized precision 0.71 under the corrected rule (0.38 for Sonnet 5), is released beside it. The rerun’s flags are the corrected rule’s verdicts, whose precision is 78 percent by the hand reading of Section 6 (71 to 84), so Figure 2’s π is nominal and its arm at $\pi = 1$ sits nearer 0.8. Every scored round-0 quotation was traced to the final draft (Table 2).

Satisfying the checker by deletion. Under true feedback the final strict rate runs from 0.092 for Qwen3-30B to 1.000 for Sonnet 5 (Table 2). Sonnet 5 had 35 flagged quotations at round 0: 1 corrected in place, 2 replaced by an accurate quotation from the same case, 2 dequoted, 30 deleted, while only 4 of its 69 accurate quotations went, against 21 under revision with no feedback. Qwen3-30B left most flags standing and Mistral Small nearly half; the other five removed most of what was flagged. Dequoting is not deletion: the words stay, the citation stays in 100 of 103 cases, and dropping the marks is the right repair when the passage is a paraphrase, the dominant residual at the top. Counting dequoting as repair, deletion still exceeds it for 6 of seven models, Grok 4.3 excepted (23 dequoted of 37, 14 deleted). Counted over all 24 episodes, accurate quotations per draft rose for 1 of seven models (Mistral Small, 0.58 to 0.75) and were flat or lower for the rest (Sonnet 5 2.88 to 2.83), so the loop raised the rate over surviving quotations and not the number of correct ones in a draft. Existence rose by the same route: of 113 citations unresolved at round 0, 82 were gone from the final draft. A *passage* arm put the evidence

in the flag: each quotation line carried the closest passage of the cited opinion. Repair rose in six of seven models, Grok 4.3 excepted, to 13 of 35 for Sonnet 5 and 27 of 92 for Mistral Small against 4, and deletion alone still exceeded repair and dequoting together in 4 of seven.

Controls: feedback content and verifier precision. Without feedback (Figure 2) Sonnet 5’s strict rate still rose (0.743 to 0.796), and true feedback added 0.20 over revision alone (paired over the 23 matters scored in both arms; interval 0.11 to 0.31, $p=0.003$), as for DeepSeek (0.39, $n=12$); the differences for the other five are within noise. Sonnet 5’s correct quotations removed rise as precision falls, 4, 9, 21, 28, and 43 of 69 at nominal π of 1, 0.75, 0.5, 0.25, and 0, against 21 with no feedback. Realised precision is below the label, since the checker’s flags are right 78 percent of the time (Section 6), and its accurate side was audited at 0 of 40 wrong. The loop stops on a clean draft, so the arms differ in rounds executed too, 2.47 under true feedback against 3.21 under false, and the series is monotone in both. Holding rounds fixed keeps it: removals per episode-round run 0.048, 0.090, 0.145, 0.150, 0.200 against 0.109 with no feedback, and after one round 10 to 32 percent of the 181 accurate quotations are gone against 23. No feedback sits between π of 0.75 and 0.5 on both bases, interpolating to 0.56 and 0.66, or 0.44 to 0.52 realised, an upper bound since 78 percent is one. Below it a verifier destroys more correct work than revision alone; the arms move recall with precision, so this bounds a loop of this shape, not precision alone. Under all-false feedback five of seven models ended below their starting rate while flagged counts fell in every model: what changed was the deletion of correct quotations.

6. Residual Errors at the Top of the Range

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s text, 68 percent of Sonnet 5’s (169 of 248) are paraphrases of the correct case presented as verbatim quotation, 56 match nothing substantial in it, and 23 cite authority outside the U.S. Reports.

Real language, wrong case. Searching every inaccurate quotation against the 62,049 U.S. Reports opinions in the Caselaw Access Project cache locates 317 whose language is verbatim in an opinion other than the one cited (134 with the true source cited in the same draft, so a ceiling). By provenance, Sonnet 5’s 248 partition into 72 verbatim elsewhere, 130 found nowhere, 40 stock phrases, 6 present in the attributed case after all. A hand reading of 160 inaccurate verdicts drawn at random across models, under the rule before two later corrections that remove only checker-side verdicts, found 35 on the checker’s side (22 percent, 95 percent interval 16 to 29: non-case material in quotation marks, or a quotation bound to the wrong case), 111 the model’s, 4 failing only on a bracketed alteration, and 10 open. The share was 8 of 28 at baseline, lowest under the combined condition (4 of 42) and highest under the quota (11 of 30); reclassifying every inaccurate verdict as unpaired at its condition’s share (200 draws), 7 of the 8 survivors hold in every draw and Qwen3-30B’s temporal quotation fall in 82 percent. The count K_d is insulated, since reclassifying an inaccurate verdict as unpaired leaves the accurate count untouched; the year-adjusted fit is not, and is exploratory.

Right case, wrong opinion. The archives hold opinion, concurrences, dissents, and head matter in one record, so a dissent’s words score accurate against the case, which is

its own misrepresentation. Of the 1,578 accurate quotations the Caselaw Access Project’s markers place (73 percent of those scored accurate), 39 sit outside the Court’s opinion (20 concurrence, 11 dissent, 8 syllabus) and 33, 2.1 percent of the placed, carry no signal in the draft.

Right case, wrong page. Of the 9,129 pinpoints, 73 percent were checked against the cited case’s page span (the rest cite other reporters or cases past volume 572) and 1,001 carry a quotation the case contains somewhere, the page-level rate’s scope; for those (42 to 262 per model) the quotation is on a cited page 85 percent of the time for Sonnet 5, 52 to 67 for the other six, 90 for the briefs, and on an adjacent page 9 percent.

7. Discussion

Where the checker belongs. Two recommendations follow, both inferences, since no arm had a person in it. The first is the checker at the gate over a loop resolving bare flags: removing an unverifiable quotation answers a flag defensibly, false flags get the same answer, and the tool still reports a clean draft.

Verifier precision is a deployment requirement. Our checker’s error is 22 percent by the hand reading, and published detectors span this axis: on one benchmark the highest-recall agent reaches 84.4 percent recall at 40.8 percent precision, the highest-precision one 76.1 percent at 62.8 percent recall [3], so the first sits below the crossing above and the second above it. A pipeline feeding a detector’s flags to a model should report precision beside recall. The second is to report the strict standard as defined here rather than the 0.85-coverage rate, which admits one wrong word in seven.

What evaluations should include. The observed rate is $R_d = K_d/|S_d|$, and a model can raise it by shrinking what it exposes, $|S_d|$, as well as by raising the correct output K_d : under pressure the strongest model quoted less, under feedback every model deleted what was flagged, $|S_d|$ down and K_d flat or lower in both. A rising rate does not imply more correct output; an evaluation scoring only what a system emits should report K_d beside R_d .

8. Limitations

Both tasks are United States federal case law, the stakes clause names a rule governing neither forum, and the checks stop at existence, wording, and page: three model panels for relevance, calibrated against the expert misrepresentation labels of Liu et al. [3], fell short of a pre-declared 0.75 bar [29], so whether a quotation supports its proposition, or the case is still good law, is untested. Everything after the primary fit is exploratory. The readings behind the checker’s error are a single unblinded pass without a second rater and predate two corrections that only remove checker-side verdicts, so 78 percent is a lower bound. The between-template spread (15 points) exceeds many single-run effects, the models are one per vendor, and capability is indexed by baseline existence, an ordering that differs on other axes.

9. Conclusion

The failure mode moves with capability: the 30B model loses existence, the middle loses quotation accuracy and correct quotations per draft, and the strongest model’s rate holds only because it quotes less; returned to it, the checker is satisfied by removal.

<https://github.com/hardy30894/citation-under-pressure>

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